INTEGRATED LEARNER

“NATION’S SECURE AI POWERED STUDENT COMMUNITY PLATFORM THAT EMPOWERS & BUILDS CONFIDENCE THROUGH COLLABORATIVE LEARNING”

SEMESTER – I PROJECT (PRD)

Product Vision & Mission

Vision: “To Ensure a Safe & Trusted digital platform that enables access & provides opportunity for every university student in nation to learn, collaborate and grow efficiently in achieving goals.”

Mission: “To Encourage Customized social & academic collaboration for student well being with seamless study and peer group matching, simple communication within a single digital system.”

Executive Summary

Integrated Learner is an ‘Unified AI driven Collaborative Platform’ for the University students who are struggling with fragmented communication and helps to form a bridge between students academic and social lifestyle by building structured peer coordination that matches specific learner compatible with study groups and partners.

The primary Business Goal is to ensure AI assisted coordination for University Student Ecosystem through a phased roll out, beginning with Tier – 1 Indian cities and gradually expanding to Nationwide Community.

The key features include:

- Building Social Relations & Supportive Campus Environment.
- Reduces Low Academic Performance.
- Single Source of Community Engagement.
- Personalized Learning & Accessibility.

Problem Statement

“When University Students attempt to manage their social and academic lives, they need a trusted and secured way to find academic peer community that suites and stay informed about supportive campus activities but struggle because of multiple unstructured collaborative platforms leading to Misinformation and Disengagement.”

User Persona

Name: Simran Saigal

Age: 21

Location: Mumbai

Education: B.Tech in Computer Science, 3rd Year

Bio: Simran, A Tech Savvy student, likes to form group with friends but struggles hard to manage studies and group work with many different apps. She wants a simple and safe space to connects with the peers that match and deserve better.

Gains & Motivations:

- Ease of Study & Collaborate.
- Safe & Trusted Digital platform.
- Clear, Calm & Well organized learning.

Pains & Frustations:

- Uses various disconnected apps for studies.
- Faces coordination and communication challenges.
- Concerned about data misuse & security.

Quote: “I just want a simple & centralized platform for all’ my study groups & campus updates that helps me to align me with desired commitments.”

Market Sizing

Using Top – Down Approach, basis of secondary research data should be estimated as follows:

- Indian population is 140 crores or 1464 million.
- Indian Total graduate student population is 4.8 crores or 48 million.
- Indian Total Tier -1 city graduate student population is 0.55 crores or 5.5 million.
- Indian Top 4 Tier -1 city graduate student population is 0.3 crores or 3 million. (30 lakhs)
- Consider (Mumbai, Bangalore, Hyderabad &Delhi) Graduate Student Top 4 Tier -1 cities population for Phased Roll out and further expanding across entire Tier – 1 cities and PAN India Coverage, Average Population per each city is 0.75 million or 7.5 lakhs.

Estimated smartphone usage among all Tier – 1 city students contributes to 85%(approx.)

- Total smartphone users (all 8 Tier – 1 cities) = 5.5*0.85 = 4.675 Million.
- Smartphone users (Top 4 Tier – 1 cities) = 3*0.85 = 2.55 Million (54.54%).
- Total percent Contribution of Top 4 Tier -1 cities for smartphone usage = 2.55/4.675*100 =54.54%

This 2.55 Million is Market Coverage(Portion of SAM) or MVP reachable geographically.

Expected Early Adoption to be 10 – 15 % then:

- It accounts around 2.5 to 4 Lakhs potential users which is our capturable market size or Serviceable obtainable Market(SOM).

Solution Overview (MVP)

The AI driven Study Group Matchmaker is the Core feature of the MVP that address and solves the challenges of university students under a single roof with a balanced approach and its working derives the key features as:

- **Student Onboarding & Profile setup –** Ensure Safe & Trusted community with university verified single sign on (SSO) Login.
- **AI Matchmaking Engine –** Create study groups by carefully comparing profiles based on subject interests, schedule overlap that fits best using AI recommendation model.
- **Group Creation & Communication –** Foster engagement & coordination through well structured peer group chats, social & academic updates over unorganized chat groups scattered across multiple apps.
- **Feedback System –** Receiving students rating in terms of satisfaction and usefulness and Iterating the loop helps to fine tune AI model to cope up with future uncertainities and enables further improvements to be deployed if needed and required.

Prioritization Framework

To Justify and confirm with the proposed MVP feature set at its Initial Launch, I strongly feel to be prioritized using MoSCoW model as:

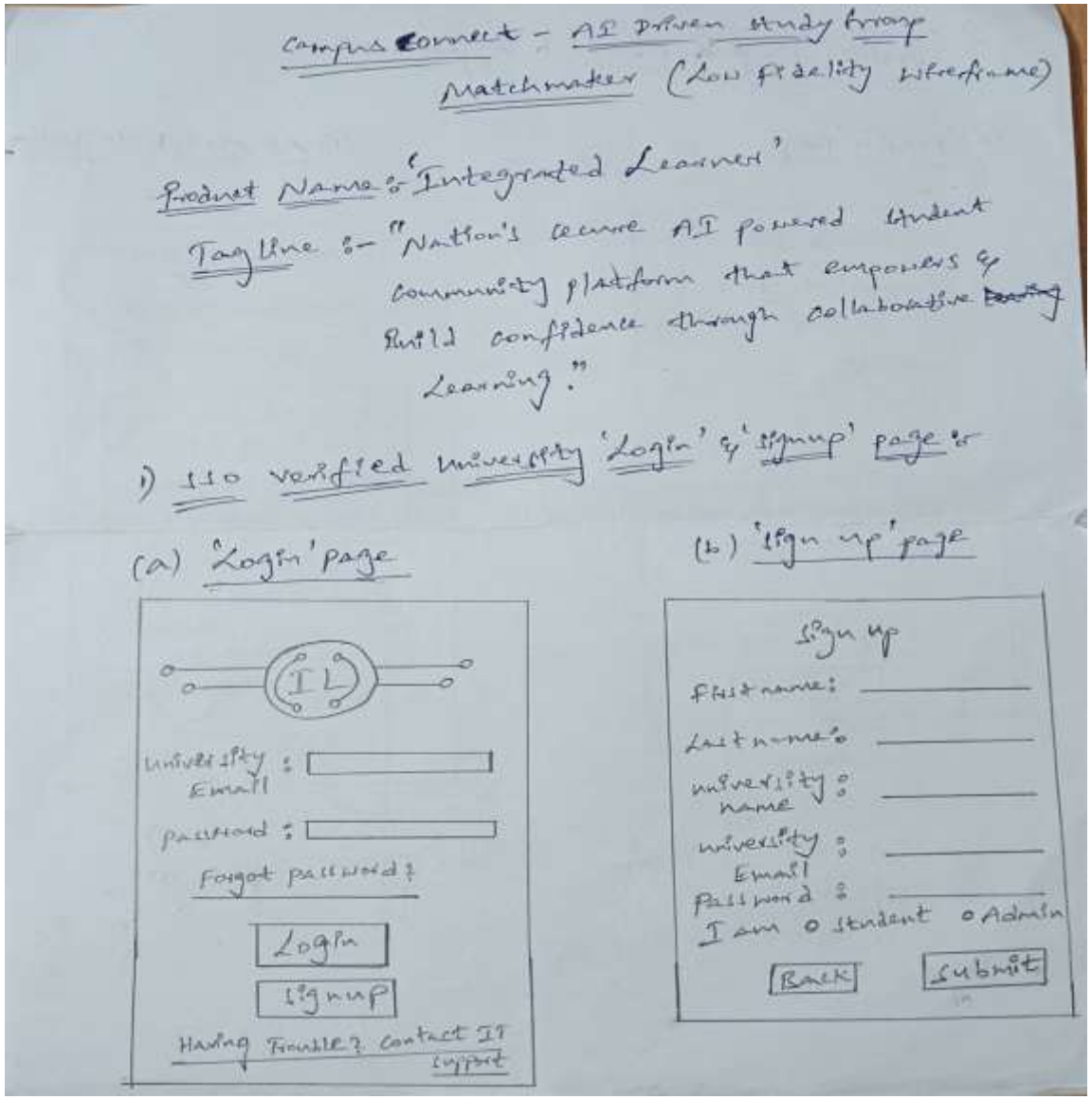
- **Must Have:** Student Onboarding & Profile set up, AI Matchmaking Engine.
- **Should Have:** Group creation & Communication, Feedback System.
- **Could Have:** Campus Event Board (remind about upcoming events).
- **Won’t Have:** Gamification & Analytics (To add points, badges and Track usage & engagement – Ideal to launch post MVP launch).

❖ Initially, The MVP Launch is planned for Top 4 Tier - 1 Cities and then expanding across entire Tier 1 as a whole and Nation wide for a Phased Roll Out in Future.

User Stories & Acceptance Criteria

Features	User Story	Acceptance Criteria
1. Student Onboarding & Profile setup	"As a University Student, I want a secure & verified single sign on (SSO) using my University email and complete academic profile, So that I can have access to AI driven space with compatible group matches that builds Trust & Confidence ".	1. "Given I use my verified University email, When I login through SSO for the first time, Then I need to be granted secure access to complete my academic profile setup before proceeding". 2. "Given I have entered valid academic details, When I save my profile, Then the data has to be stored successfully enabling AI powered study group matches".
2. AI Matchmaking Engine	"As a Student, I want a structured AI recommendation model that suggests study groups based on my interests, So that I can connect to exact peer groups that deserves with ease".	1. "Given my profile setup is complete, when I clicked on 'find matches', then AI should suggest me with appropriate suitable suggestions". 2. "Given I select a match, When I clicked on 'Join group', Then I should be added to group's chat".
3. Group Creation & Collaboration	"As a Group Member, I want to chat & share study updates, So that our group can coordinate easily".	1. "Given I'm part of group, When I want to send a message, Then the message should be visible to all' the Group Members Instantly". 2. "Given the Chat is Active, When I send a file in the Group, Then It should be Uploaded Successfully".
4. Feedback System	"As a Student, I want to rate my Study Group Experience, So that AI can refine its model to provide Relative Suggestions for Future Matches".	1. "Given I Complete a Study Session, When I access the Feedback Page, Then there should be a Simple & Intuitive Visual Rating Form". 2. "Given I Submit a feedback, When I press 'Submit', Then I should see a Confirmation Dialog".
5. Campus Event Board	"As a Student, I want to informed with Updates, So that I Stay Tuned with the Group's Activity".	1. "Given I'm in Active Group, When a member send a Message, Then I should receive Notification". 2. "Given I Mute Notification, When new message arrives, Then I should not be Distracted".

Low Fidelity Wireframe (Rough Sketch)



2) 'Profile' page & AI Match suggestions :-

(a) 'profile' page

Profile setup

'personalise your experience'

User ID/Email : _____

Course : _____

Study Time : _____

Preferred Mode : _____

Skills : _____

City : _____

Submit

(b) AI Match suggestions

AI Match suggestions

'Explore peer groups you deserve'

○ Peer : Simran Saigal
Btech CSE | 81% Match
[Common Subs: All] [Study time :-]
Join Group

○ Study group : Data structures
Btech CSE | 92% Match
[Common Subs: DS] [Study time :-]
Join Group

Refresh suggestions

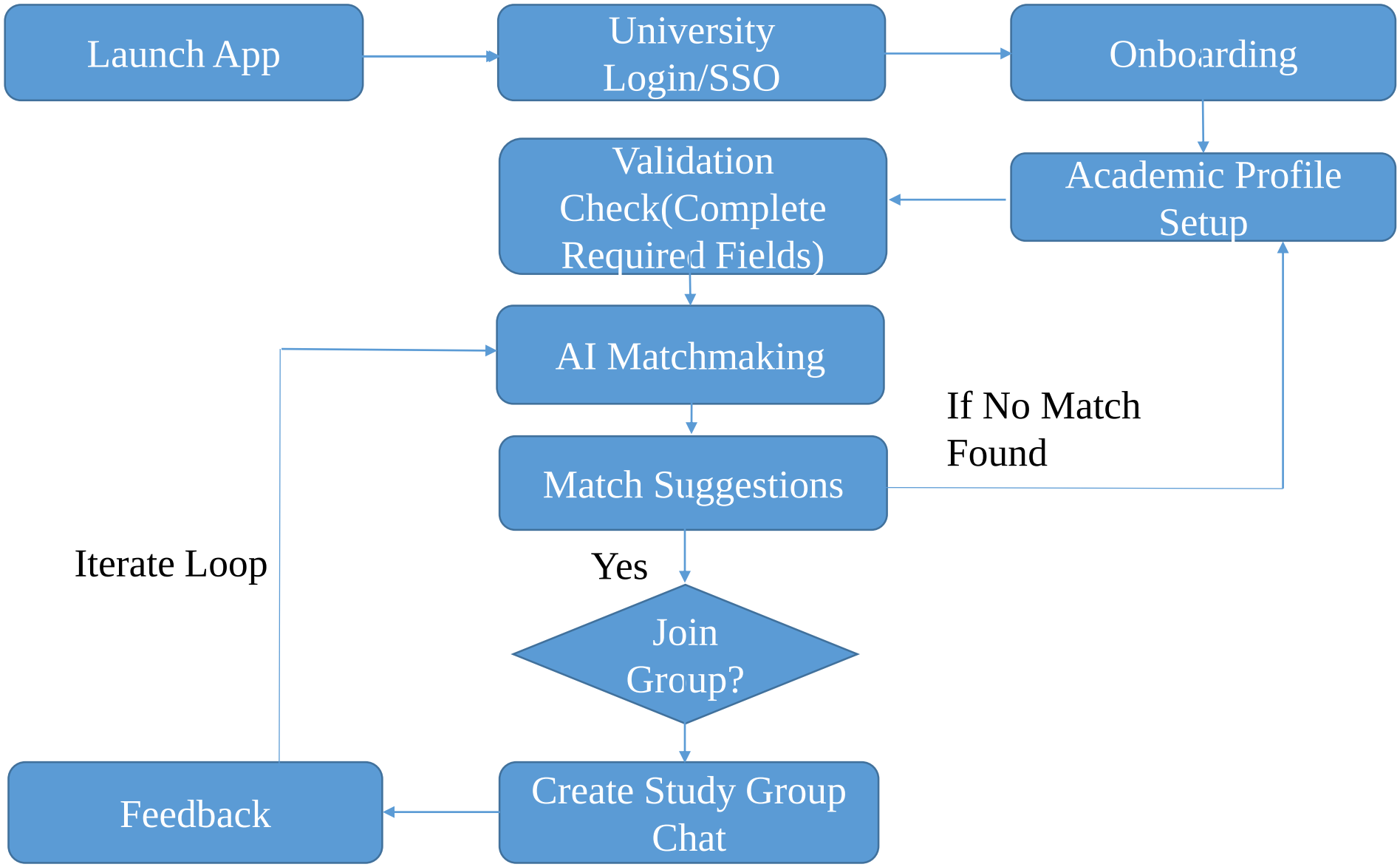
Back

3) 'Dash board' & 'feedback' page :-

(A) Dash board

(B) feed back

User Flow Diagram



Technical Requirements

AI Input Parameters	AI Output Parameters	Outcome
1. Subject/Course Interests	Curated list of peers with similar Interests.	Validate Academic Compatibility.
2. Preferred Study Time	Suitable groups with Mutual Availability.	Improves coordination & Participation.
3. Learning Style	Community Suggestions aligned with learning patterns.	Ensures Smooth Collaboration.
4. Feedback Rating	Iterate and Refine model for future appropriate matches with valid suggestions.	Measures Satisfaction.

The Third Party API Integrations include:

❖University Authentication API (SSO) – Verifies Student Identity through University email Login.

❖Cloud Database – Stores profiles, Chat logs & Group Data.

❖AI Recommendation Engine – Runs Similarity Matching Algorithm.

❖Messaging API – Enables Group Chats & Notifications.

Launch Success Metrics (KPIs)

AARRR Funnel Stage	KPI Metric	Quantifiable Success Metrics (Expected)
Acquisition	Verified User Signups	1000 verified signups Monthly.
Activation	Profile completion & AI Match Engagement	60% of the onboarded students with relevant matches weekly.
Retention	Active Study Group Participation	50% weekly active engagement.

Conclusion

Integrated Learner is a digital platform that envisages creating a safe, intelligent and collaborative academic ecosystem, Providing access for the university students to bring equilibrium in their social & academic Frustated lives through structured AI Generated Matchmaking Model that builds trust. At its Inception during MVP Launch, the Product aims at Top 4 Tier – 1 Indian cities for quick response and focused targeting, thereby progressively expand across countrywide scale through phased roll out.

References

- https://www.watermarkinsights.com/resources/blog/the-top-6-challenges-college-students-face/
- https://amberstudent.com/blog/post/15-most-common-problems-faced-by-students-and-their-solutions